

^{36}K εp decay (341 ms) 1996II02,1980Es01

Parent: ^{36}K : $E=0$; $J^\pi=2^+$; $T_{1/2}=341$ ms 3; $Q(\varepsilon\text{p})=4307.4$ 3; $\% \varepsilon\text{p}$ decay= 5.1×10^{-2}

^{36}K - J^π , $T_{1/2}$: From the Adopted Levels of ^{36}K (2012Ni01).

^{36}K - $Q(\varepsilon\text{p})$: From 2021Wa16.

^{36}K - $\% \varepsilon\text{p}$ decay: $\% \varepsilon+\beta^+\text{p}=5.1 \times 10^{-4}$ is scaled by evaluators from $\% \varepsilon+\beta^+\text{p}=4.9 \times 10^{-4}$ reported in 1996II02. $\% \varepsilon+\beta^+\text{p}=4.9 \times 10^{-4}$ in 1996II02 was normalized based on the strongest 970-keV $\varepsilon+\beta^+$ -delayed proton $I_p=3.3 \times 10^{-4}$ 9 measured by 1980Es01. 1980Es01 reported $\% \varepsilon+\beta^+\text{p}=4.8 \times 10^{-4}$ and their I_p was normalized based on the strongest 1970-keV $\varepsilon+\beta^+$ -delayed γ ray $\% I_\gamma=79$ 8 measured by 1972Mi13. Evaluators adopt a more precise $\% I_\gamma(1970)=81.8$ 16 from the evaluated ^{36}K ε decay dataset (2012Ni01,1976Fr03) as the new normalization reference, and therefore, all original I_p values from 1996II02 and 1980Es01 are scaled by 1.04 11.

1996II02,1997II03: a 500-MeV proton beam was produced from the TRIUMF cyclotron and bombarded a CaO powder target. The spallation reaction products diffused out of the target and were selectively ionized at a Re surface. $A=36$ singly charged ions were selected by a magnetic mass analyzer and implanted into a carbon foil at 7.0×10^5 pps. Protons and α particles were detected using a 30- μm silicon surface barrier detector with FWHM=23 keV. ^{32}S and ^{35}Cl recoils were detected using an MCP in a back-to-back geometry for timing stop signals. Measured $E_p(>0.5$ MeV) and I_p for 15 proton peaks, $E_\alpha(>1.5$ MeV) and I_α for 19 α peaks, p(recoil)-coin, and α (recoil)-coin. Deduced levels, J , π , decay branching ratios, $\log ft$, partial decay widths for ^{36}Ar , and ratios of $^{35}\text{Cl}(p,\alpha)^{32}\text{S}$ to $^{35}\text{Cl}(p,\gamma)^{36}\text{Ar}$ reaction rates.

1980Es01: a 20-MeV proton beam was produced from the University of Jyväskylä MC-20 cyclotron and bombarded a stack of four ^{36}Ar -implanted Al foils. The $^{36}\text{Ar}(p,n)$ reaction products were thermalized by helium, transported with NaCl-loaded helium, and deposited onto a fast-transport Al-coated mylar tape in a measurement chamber. Protons and α particles were detected using 14 and 26- μm fully depleted (FWHM=40 keV) and 100- μm partially depleted Si(Au) surface barrier detectors. A polyethylene absorber foil was placed in front of Si(Au) to cause an energy shift for distinguishing p/ α . Measured $E_p(>0.5$ MeV) and I_p for 11 proton peaks, $E_\alpha(>2.0$ MeV) and I_α for 12 α peaks. Deduced levels, J , π , decay branching ratios, $\log ft$, and partial decay widths for ^{36}Ar .

1980Ew01: a 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC_2 target. The $^{45}\text{Sc}(p,7n3p)$ spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a carbon foil for proton measurements. Protons and α particles were detected using 20- μm and 100- μm silicon surface barrier detectors with FWHM=35 and 23 keV. Measured $E_p(>0.5$ MeV) and I_p for 3 proton peaks, E_α and I_α for 4 α peaks. Deduced levels, J , π , and decay branching ratios for ^{36}Ar .

 ^{35}Cl Levels

<u>E(level)</u>	<u>$J^\pi^\dagger$</u>
0	$3/2^+$

† From the Adopted Levels.

Delayed Protons (^{35}Cl)

<u>E(p)†</u>	<u>E(^{35}Cl)</u>	<u>I(p)†@</u>	<u>E(^{36}Ar)$^\#$</u>	<u>Comments</u>
501.8 11	0	1.14×10^{-6} 33	9023	E(p): other: 501 10 (1980Es01). I(p): other: 3.6×10^{-6} 13 (1980Es01).
622.7 14	0	1.6×10^{-6} 5	9147	
693.5 8	0	7.9×10^{-5} 22	9220	E(p): other: 693 5 (1980Es01), 691 8 (1980Ew01). I(p): other: 7.8×10^{-5} 24 (1980Es01).
851.7 10	0	2.3×10^{-5} 7	9383	E(p): other: 849 5 (1980Es01). I(p): other: 2.0×10^{-5} 7 (1980Es01).
969.6 12	0	3.4×10^{-4} 10	9504	E(p): other: 970 5 (1980Es01), 959 7 (1980Ew01). I(p): from 1980Es01 and also adopted as the normalization reference by 1996II02.
1168.5 14	0	6.2×10^{-7} 7	9709	

Continued on next page (footnotes at end of table)

^{36}K ε p decay (341 ms) [1996II02,1980Es01](#) (continued)Delayed Protons (continued)

$E(p)^{\dagger}$	$E(^{35}\text{Cl})$	$I(p)^{\dagger@}$	$E(^{36}\text{Ar})^{\#}$	Comments
1198.5 11	0	2.5×10^{-6} 7	9740	
1272.0 20	0	1.6×10^{-6} 5	9815	
1334.2 7	0	4.5×10^{-5} 12	9879	E(p): other: 1333 5 (1980Es01), 1325 7 (1980Ew01). I(p): other: 3.6×10^{-5} 11 (1980Es01).
1409.3 22	0	1.04×10^{-6} 33	9957	
1874 ‡ 10	0	1.1×10^{-6} 5	10435	
1992 ‡ 10	0	5.0×10^{-6} 20	10556	
2048 ‡ 10	0	4.9×10^{-6} 19	10613	
2458 ‡ 10	0	3.0×10^{-6} 13	11035	
2640 ‡ 10	0	2.1×10^{-6} 10	11222	

† From [1996II02](#), unless otherwise noted. E(p) is in lab frame.

‡ From [1980Es01](#) and also adopted by [1996II02](#).

$^{\#}$ $E(\text{level})(^{36}\text{Ar}) = E(p)_{\text{lab}} \times [m(p) + m(^{35}\text{Cl})] / m(^{35}\text{Cl}) + S(p)(^{36}\text{Ar}) + E(\text{level})(^{35}\text{Cl})$, where $S(p)(^{36}\text{Ar}) = 8506.97$ 4 ([2021Wa16](#)). Proton decay is assumed to populate only the ground state of the daughter nucleus ^{35}Cl based on the arguments proposed in [1996II02](#): (A) unreasonably small $\log ft$ values if populating excited states; (B) no pairs of particle groups characteristic of the g.s./first excited state energy difference; (C) compatible level matching with resonance reaction studies.

$^@$ Absolute intensity per 100 decays.

^{36}K εp decay (341 ms) 1996I102,1980Es01Decay Scheme

I(p) Intensities: I(p) per 100 parent decays

